In [1]:
```
import pandas as pd
import numpy as np
```

In [2]:
```
data=pd.read_csv(r"C:\Users\bhuva\Downloads\Automobile.csv")
```

In [3]:
```
data
```

Out[3]:

	name	mpg	cylinders	displacement	horsepower	weight	acceleration	model
0	chevrolet chevelle malibu	18.0	8	307.0	130.0	3504	12.0	
1	buick skylark 320	15.0	8	350.0	165.0	3693	11.5	
2	plymouth satellite	18.0	8	318.0	150.0	3436	11.0	
3	amc rebel sst	16.0	8	304.0	150.0	3433	12.0	
4	ford torino	17.0	8	302.0	140.0	3449	10.5	
...	...	...	...	...	...	...	...	
393	ford mustang gl	27.0	4	140.0	86.0	2790	15.6	
394	vw pickup	44.0	4	97.0	52.0	2130	24.6	
395	dodge rampage	32.0	4	135.0	84.0	2295	11.6	
396	ford ranger	28.0	4	120.0	79.0	2625	18.6	
397	chevy s-10	31.0	4	119.0	82.0	2720	19.4	

398 rows × 9 columns

In [5]:
```
data.shape
```

Out[5]:
```
(398, 9)
```

In [9]:
```
data.columns
```

Out[9]:
```
Index(['name', 'mpg', 'cylinders', 'displacement', 'horsepower', 'weight',
       'acceleration', 'model_year', 'origin'],
      dtype='str')
```

In [11]:
```
data.dtypes
```

Out[11]:
```
name               str
mpg            float64
cylinders        int64
displacement   float64
horsepower     float64
weight           int64
acceleration   float64
model_year       int64
origin             str
dtype: object
```

In [12]:
```
data.describe
```

Out[12]:
```
<bound method NDFrame.describe of                          name   mpg  cylinde
rs  displacement  horsepower  \
0    chevrolet chevelle malibu  18.0          8         307.0       130.0
1            buick skylark 320  15.0          8         350.0       165.0
2           plymouth satellite  18.0          8         318.0       150.0
3                amc rebel sst  16.0          8         304.0       150.0
4                  ford torino  17.0          8         302.0       140.0
..                         ...   ...        ...           ...         ...
393            ford mustang gl  27.0          4         140.0        86.0
394                  vw pickup  44.0          4          97.0        52.0
395              dodge rampage  32.0          4         135.0        84.0
396                ford ranger  28.0          4         120.0        79.0
397                 chevy s-10  31.0          4         119.0        82.0

     weight  acceleration  model_year  origin
0      3504          12.0          70     usa
1      3693          11.5          70     usa
2      3436          11.0          70     usa
3      3433          12.0          70     usa
4      3449          10.5          70     usa
..      ...           ...         ...     ...
393    2790          15.6          82     usa
394    2130          24.6          82  europe
395    2295          11.6          82     usa
396    2625          18.6          82     usa
397    2720          19.4          82     usa

[398 rows x 9 columns]>
```

In [14]:
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 398 entries, 0 to 397
Data columns (total 9 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   name          398 non-null    str
 1   mpg           398 non-null    float64
 2   cylinders     398 non-null    int64
 3   displacement  398 non-null    float64
 4   horsepower    392 non-null    float64
 5   weight        398 non-null    int64
 6   acceleration  398 non-null    float64
 7   model_year    398 non-null    int64
 8   origin        398 non-null    str
dtypes: float64(4), int64(3), str(2)
memory usage: 35.9 KB
```

In [15]:
```
data.isnull().sum()
```

Out[15]:
```
name            0
mpg             0
cylinders       0
displacement    0
horsepower      6
weight          0
acceleration    0
model_year      0
origin          0
dtype: int64
```

In [16]:
```
missing_percentage = (data.isnull().sum() / len(data)) * 100

missing_percentage
```

Out[16]:
```
name            0.000000
mpg             0.000000
cylinders       0.000000
displacement    0.000000
horsepower      1.507538
weight          0.000000
acceleration    0.000000
model_year      0.000000
origin          0.000000
dtype: float64
```

In [17]:
```
median_horsepower = data["horsepower"].median()

data["horsepower"] = data["horsepower"].fillna(median_horsepower)
```

In [18]:
```
data.isnull().sum()
```

Out[18]:
```
name            0
mpg             0
cylinders       0
displacement    0
horsepower      0
weight          0
acceleration    0
model_year      0
origin          0
dtype: int64
```

In [20]:
```
data.duplicated().sum()
```

Out[20]:
```
np.int64(0)
```

In [21]:
```
data = data.drop_duplicates()
```

In [22]:
```
data.columns = data.columns.str.strip().str.lower().str.replace(" ", "_")
```

In [23]:
```
data["name"] = data["name"].str.strip().str.lower()

data["origin"] = data["origin"].str.strip().str.lower()
```

In [24]:
```
numeric_columns = [
    "mpg",
    "cylinders",
    "displacement",
    "horsepower",
    "weight",
    "acceleration",
    "model_year"
]

data[numeric_columns] = data[numeric_columns].apply(
    pd.to_numeric,
    errors="coerce"
)
```

In [25]:
```
data[numeric_columns].isnull().sum()
```

Out[25]:
```
mpg             0
cylinders       0
displacement    0
horsepower      0
weight          0
acceleration    0
model_year      0
dtype: int64
```

In [26]:
```
data[numeric_columns].describe()
```

Out[26]:

	mpg	cylinders	displacement	horsepower	weight	acceleration	m
count	398.000000	398.000000	398.000000	398.000000	398.000000	398.000000	3
mean	23.514573	5.454774	193.425879	104.304020	2970.424623	15.568090	
std	7.815984	1.701004	104.269838	38.222625	846.841774	2.757689	
min	9.000000	3.000000	68.000000	46.000000	1613.000000	8.000000	
25%	17.500000	4.000000	104.250000	76.000000	2223.750000	13.825000	
50%	23.000000	4.000000	148.500000	93.500000	2803.500000	15.500000	
75%	29.000000	8.000000	262.000000	125.000000	3608.000000	17.175000	
max	46.600000	8.000000	455.000000	230.000000	5140.000000	24.800000	

In [27]:
```
print(data["origin"].value_counts())

print(data["cylinders"].value_counts())

print(data["model_year"].value_counts().sort_index())
```

```
origin
usa       249
japan      79
europe     70
Name: count, dtype: int64
cylinders
4    204
8    103
6     84
3      4
5      3
Name: count, dtype: int64
model_year
70    29
71    28
72    28
73    40
74    27
75    30
76    34
77    28
78    36
79    29
80    29
81    29
82    31
Name: count, dtype: int64
```

In [28]:
```
print("Dataset shape:", data.shape)

print("\nMissing values:")
print(data.isnull().sum())

print("\nDuplicate rows:", data.duplicated().sum())

print("\nData types:")
print(data.dtypes)

data.head()
```

```
Dataset shape: (398, 9)

Missing values:
name            0
mpg             0
cylinders       0
displacement    0
horsepower      0
weight          0
acceleration    0
model_year      0
origin          0
dtype: int64

Duplicate rows: 0

Data types:
name                str
mpg             float64
cylinders         int64
displacement    float64
horsepower      float64
weight            int64
acceleration    float64
model_year        int64
origin              str
dtype: object
```

Out[28]:

	name	mpg	cylinders	displacement	horsepower	weight	acceleration	model_y
0	chevrolet chevelle malibu	18.0	8	307.0	130.0	3504	12.0	
1	buick skylark 320	15.0	8	350.0	165.0	3693	11.5	
2	plymouth satellite	18.0	8	318.0	150.0	3436	11.0	
3	amc rebel sst	16.0	8	304.0	150.0	3433	12.0	
4	ford torino	17.0	8	302.0	140.0	3449	10.5	

In [30]:
```
data.to_csv("Automobile_Cleaned.csv", index=True)
```

In [ ]:
```

```